

Chapter 5 — Introduction to data cleaning

After collection and annotation, raw tables still need preparation before reliable analysis or supervised training

Learning objectives

- Distinguish cleaning from preprocessing, name four practical motives for both phases
- Recognize schema discipline when joining multiple corpora

Why cleaning matters

- Without cleaning, missing fields, duplicates, inconsistent encodings
- Cross-corpus work makes those defects visible at pipeline scale

Example 5.1 — Missing target values in classification

- Example 5.1 — hands-on module
- Example 5.1 shows a fraud table with blank labels
- Supervised training either drops those rows or invents labels
- Explore the chapter example module
- View files: ``modules/chapter5/example1/``

Example 5.2 — Dropping unused identifier columns

- Example 5.2 — hands-on module
- Example 5.2 removes high-cardinality identifiers such as raw transaction IDs that never
- Memory and training time improve without changing the predictive schema
- Explore the chapter example module
- View files: ``modules/chapter5/example2/``

Cleaning versus preprocessing

- This book treats cleaning and preprocessing as consecutive phases
- Non-model fields
- Preprocessing transforms cleaned values into representations algorithms expect

Example 5.3 — Duplicate retail transactions

- Example 5.3 — hands-on module
- Example 5.3 shows the same order ID appearing twice after a system retry
- Counting both rows inflates revenue and customer frequency until duplicates are removed or
- Cleaning precedes the scaling and encoding examples introduced later in the chapter
- View files: ``modules/chapter5/example3/``

Four recurring motives

- Four motives recur
- Detailed failure modes appear in the next parts; remedies follow in Sections 5.3 and 5.4

Takeaways

- Cleaning makes values trustworthy; preprocessing makes them algorithm-ready
- Both follow annotation and precede exploration-heavy modeling work
- Schema alignment across sources is itself a cleaning task

Next

- Complete the quiz for this part
- Continue to missing data, MCAR, MAR, MNAR mechanisms and their causes